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A DoS Attack Against 5G-Based IIoT Networks Utilizing Standard Network Slicing Mechanisms

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ABSTRACT 5G networks promise to integrate several new technologies including cloud-based virtualization of physical resources and network functions, as well as the separation of the control plane from the data plane. Network slicing – the ability to have multiple virtual networks simultaneously share the same physical infrastructure – plays a central role in these technologies. Network slicing, however, opens the door to new attacks from malicious network tenants. In this work, we show how a side-channel that leaks information about one network tenant to another can be created and used to launch a Denial of Service Attack (DoS). The attack involves three main steps: a reconnaissance phase to obtain the victim network's traffic patterns over time by exploiting a side channel based on standard 5G network slicing mechanisms; training a Long Short-Term Memory (LSTM) model to be able to predict the victim network's peak traffic times; and launching the DoS attack during those peak periods, again via standard 5G network slicing mechanisms. A successful attack on a typical Industrial IoT (IIoT) smart manufacturing application was demonstrated using OMNeT++ simulations and real IoT-23 traffic traces.

INDEX TERMS 5G Networks; Network Slicing; DoS Attacks; Side-Channel Attacks; LSTM Model; OMNeT++ Simulations.

I. INTRODUCTION

5G networks have ushered in a new era of connectivity by embracing cloud-native core services [1]. Unlike previous generations, they integrate Software-Defined Networking (SDN), Service-Based Architecture (SBA), and Virtualized Network Functions (VNF) to achieve unprecedented flexibility and scalability [2]. SDN decouples the control plane from the data plane, enabling centralized management and dynamic traffic routing [13-14]. SBA facilitates seamless orchestration across diverse network elements [15], while VNF abstracts network functions from underlying hardware [16]. These innovations allow for unprecedented flexibility and scalability, enabling dynamic provisioning of network resources through network slicing [3-5].

Network slicing a cornerstone of 5G partitions a single physical infrastructure into multiple virtual networks, each calibrated for specific service requirements [6-7]. This capability maximizes resource utilization and accommodates use cases ranging from enhanced mobile broadband (eMBB) to ultra-reliable low-latency communications (uRLLC) and massive machine-type communications (mMTC) [8-9]. Yet,

the shared nature of these logical slices introduces new vulnerabilities that traditional security mechanisms struggle to address [10–12]. A critical vulnerability arises when an attacker exploits slice allocation mechanisms to launch a Denial-of-Service (DoS) attack. By monitoring slice utilization, the adversary can predict peak demand windows and preemptively consume available resources. The resulting resource shortage prevents the victim slice from scaling at crucial moments, leading to service disruption [10, 17-18]. Hence, there is a need to explore potential DoS threats that exploit the standard mechanisms of 5G network slicing to be able to design proper detection and mitigation techniques.

In this work, we explore a potential side channel that arises from standard 5G network slicing mechanisms and can leak physical network utilization of one slice (including peak times) to co-resident slices, thus circumventing the slice isolation mechanisms altogether. Our threat model assumes the following:

- The attacker and victim share the same access cell (i.e. *gnodeB*),

- Standard 5G slice allocation APIs are used to request and release capacity,
- No protocol violations or compromised devices are required.

An attacker can target a specific victim network using the same access cell, and use a side-channel based on standard 5G network slicing mechanisms to estimate its traffic patterns over a period of time (e.g. a work week). The attacker then builds a machine learning model (in this case, a Long Short-Term Memory Networks (**LSTMs**)) of the victim network's traffic patterns and use it to predict the victim's peak windows during the attack. By requesting large slice allocations just before those peaks, the attacker can create network resources shortage, deprives the victim network from acquiring the slice capacity it needs, and disrupt its operation. This timing-based resource exhaustion renders the victim unable to scale its network services and degrades its quality of service.

In this work, a smart manufacturing facility with two sites is used as a simulation testbed to demonstrate such an attack. Using **OMNeT++** [50] simulation and the real traces from **IoT-23** data set [44], the study demonstrates how an attacker could predict traffic patterns of the victim network to launch a DoS attack.

Unlike previous studies that focused on generic IoT or LTE-based networks, this work specifically targets 5G network slicing environments in industrial IoT deployments, introducing a predictive DoS attack leveraging LSTM-based temporal traffic modeling. This integration has not been explored in the reviewed literature.

Researchers have demonstrated that LSTM models can be used to predict network traffic in complex and dynamically changing 5G environment [45-46]. While in [45] this was used to improve both traffic management and resource allocation, in [46], these models were used to learn the timing patterns of benign network traffic and use them to reshape malicious traffic to evade Network Intrusion Detection Systems (**NIDS**) with a high success rate.

In [47], vulnerabilities in 5G core networks were investigated by simulating DoS attacks on the Packet Forwarding Control Protocol (**PFCP**) a core signaling protocol used between SMF and UPF in the 5G architecture. Their attack demonstrates how flooding protocol-layer messages can exhaust control-plane resources and disrupt session management at the core. A DoS attack against the 5G network slicing was introduced in [48]. It requires a large number of compromised users' devices to send repeated and distributed slice mobility requests. This is basically a flooding against the network control functions themselves (i.e. operator) and not against any specific overlay network. Since the attack is not based on any specific knowledge about the network traffic at the time of the attack, it would require a large number of compromised devices and high level of coordination.

Conversely, the attack proposed in this work can target a specific 5G-based industrial IoT (**IIoT**) by preventing it

from acquiring the appropriate network slice it needs without the need of a large coordinated attack by compromised devices. Other studies [49] highlight the fact that while machine learning can significantly enhance threat detection and mitigation in 5G networks, it also introduces new vulnerabilities. Adversaries can leverage machine learning to craft sophisticated attacks that evade traditional security mechanisms.

A recent study [58] revealed vulnerabilities in the 5G protocol implementations on edge devices that allow DoS attacks. These however, are highly device-specific and focus on protocol stack bugs rather than exploiting systemic architectural behaviors like slicing. While these protocol-based DoS attacks have exposed important gaps in signaling and control-plane resilience they generally require protocol violations, system weaknesses, or large-scale device compromise. A survey of several protocol-layer threats including signaling storms, insecure handover procedures, and slice orchestration abuse is presented in [57]. While these findings are valuable, such attacks typically require extensive resources, large botnets, or depend on exploiting specific weaknesses, which limits their precision and stealth. In contrast, our approach stays within standard-compliant slicing behavior and simply takes advantage of normal resource competition to degrade service performance.

Recent works have also explored attacks at the physical layer, namely jamming [55]. Signal interference was explored to disrupt 5G New Radio (**NR**) channels. Despite their effectiveness, such attacks tend to be intrusive, often requiring high transmission power or close proximity to the target, which makes them easier to detect and harder to scale. Also, researchers showed that machine learning techniques such as XGBoost can be used to detect jamming attempts with accuracy rates above 99% [56]. Furthermore, jamming attacks typically require high power or proximity, making them easier to detect and harder to control. In contrast, our proposed method does not interact with the radio interface at all. Instead, it subtly degrades service quality by exploiting resource allocation at the slice level.

On the defense side, a specialized dataset for analyzing DoS and DDoS attacks on 5G slices was introduced in [53], and was used to train a BiLSTM-based model to detect such attacks at a success rate of 99.99%. That work however, did not explore proactive or adversarial use of learning models. Another work on detection, [54], applied deep transfer learning using CNN and BiLSTM models to identify DDoS traffic patterns. Though effective for classification, that work utilized rich and labeled datasets, it might not cover all real-world attack scenarios. Both works in [53] and [54] utilized compromised user devices on the victim networks to launch the DDoS attacks and did not target 5G network slicing mechanisms directly.

In this work, the proposed DoS attack specifically targets 5G-based IIoT networks used in smart manufacturing. IIoT

networks were singled out due to their highly regular and consistent traffic patterns (by sensors, actuators, manufacturing units etc.), with some irregular traffic (e.g. management tasks). This regularity makes possible to train a model to predict the timing of traffic peaks with high accuracy to launch a timing-based resource exhaustion attack that only uses standard 5G slice allocation interfaces to deny the victim the ability to scale up their network slice when needed the most, at peak traffic. This is different from previous DoS attacks that sought to deny the victim access to the RAN via jamming or protocol-based exploits. Furthermore, such an attack cannot be detected by firewalls or intrusion detection systems as they do not interfere in anyway with the victim network's traffic through any form of communication with the victim network or utilize a compromised node.

This work also demonstrates a new exploitable side-channel based on 5G network slicing due to shared physical network resources among logical networks. The only assumption made is that the victim network utilizes standard 5G network slicing mechanisms to optimize cost (adjust slice capacity based on traffic volume).

A complete simulation testbed has been built using OMNet++ 5G and was used to demonstrate the attack using actual IIoT traffic data set (IoT23). Next, a brief background on 5G network architecture and slicing mechanisms is presented, followed by a description of the implemented test bed. Next, detailed description of the proposed attack are presented followed by experimental results and conclusions.

II. Background

The basic 5G architecture, shown in Fig. 1, consists of four core components: Radio Access Network (RAN), Multi-access Edge Computing (MEC), Evolved Packet Core (EPC), and IP Multimedia Subsystem (IMS) [19]. It supports several use cases are such as eMBB (Enhanced Mobile Broadband), uRLLC (Ultra-Reliable and Low Latency Communications), and mMTC (Massive Machine-Type Communications). Following SDN principles, the control and data planes are separated and implemented as software components on virtualized infrastructure [16, 20, 22]. The RAN manages data transfer, while the core network handles control signaling. A Service-Based Architecture (SBA) framework is utilized [2-3] and modular functions are delivered via APIs [4]. Network Functions Virtualization (NFV) virtualizes functions traditionally tied to hardware, further enhancing SBA by running these functions as software on standard servers [1-5]. This improves resource efficiency and cost, and increases flexibility [4]. However, despite the isolation measures built into 5G network slicing, core network utilization by an overlay (i.e. virtual) network can leak into other networks, i.e. a side-channel can be exploited as will be demonstrated in this work.

A. Network Slicing

Just as elastic computing in the cloud enables flexible

resource allocation, 5G allows users to customize their network slices (NSs) to meet specific performance, security, and reliability requirements. For example, one slice can be optimized for high-bandwidth, low-latency applications like Augmented Reality, while another may be tailored for IoT deployments with low data rates. Each slice operates independently, enabling use-case-specific customization.

5G's SBA and NFV facilitate slice management by virtualizing and scaling network functions [23]. Software-Defined Networking (SDN) enables centralized control and dynamic resource allocation [24]. An end-to-end slice encompasses resources assigned across the RAN, Transport, and Core Networks. The bandwidth of the core physical network is divided into slices among different users and services. RAN slicing ensures that each slice receives appropriate radio resources, while the transport network maintains service quality across slices [25-26].

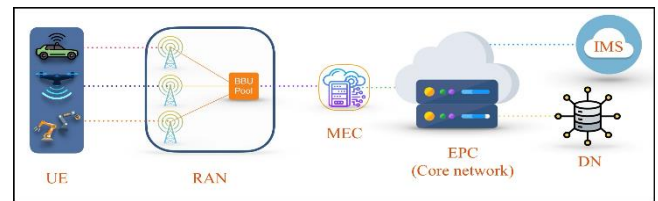


FIGURE 1. 5G network architecture.

NS Setup Process

Fig. 2 illustrates the 3GPP-standard process of setting up a network slice [29-30-33]. A user equipment (UE) sends a request with the slice's specifics (e.g., latency, bandwidth, reliability) via Non-Access Stratum (NAS) signaling, which manages control plane traffic. The RAN processes the request through the next-generation Node B (gNB), allocates the required radio resources, and forwards the request to the 5G Core. Within the core, the Access and Mobility Management Function (AMF), Session Management Function (SMF), and User Plane Function (UPF) manage the control and user planes [28]. The NS Selection Function (NSSF) selects the appropriate slice as per standard policies. The SMF sets up the session, while the UPF manages user plane traffic through the transport slice, maintaining Quality of Service (QoS) [31]. The slice, isolated from others, remains tailored to meet the UE needs, and managed by Network Slice Management Function (NSMF) and Network Slice Selection Management Function (NSSMF) [27, 32].

The time to setup a NS varies based on network configuration, load, and device capabilities. While 3GPP standards ensure efficient processes, timing specifics are typically determined by network operators to suit their infrastructure and policies [34-36]. After establishing a network slice, communication services are delivered through multiple NS Instances (NSIs).

3GPP [18, 38, 39] standards specify the following key parameters for network slicing to ensure QoS and reliability:

1. **Averaging Window:** The time period over which QoS metrics (e.g., delay, error rate) are averaged to maintain consistent performance.
2. **Maximum Data Burst Volume:** The maximum volume of data that can be transmitted as a burst without violating QoS constraints.
3. **Packet Delay Budget:** The maximum end-to-end delay of any packet.
4. **Packet Error Rate:** The acceptable error rate for data transmission reliability.
5. **Priority Level:** Determines the urgency of handling different data traffic types.
6. **Resource Type:** Specifies whether resources are guaranteed or shared, impacting service stability.

Combinations of these parameters correspond to predefined characteristics and are assigned unique 5G QoS Identifiers (5QI) as illustrated in Table I. This framework enables 5G networks to offer customized slices for a wide range of applications – from low-latency automation to high-bandwidth streaming [36, 38, 40].

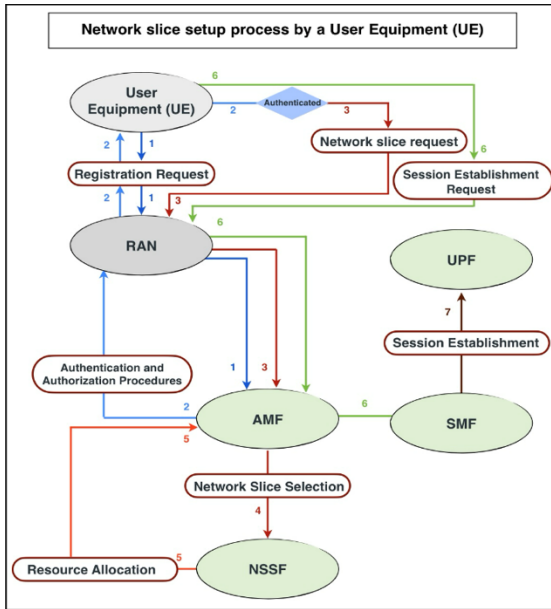


FIGURE 2. Process of setting up a network slice by a UE.

3GPP standards [29] also specify additional network slice characteristics, including:

1. **Defined Capacity:** The maximum bandwidth, latency, and performance metrics a network slice can support to satisfy a service level agreement (SLA). If the network controller cannot fulfill a slice request due to insufficient available capacity, the request is denied. This denial can inadvertently leak information about core network utilization at that moment to the requester, whose request was turned down. In such cases, complete network slice

isolation cannot be maintained [35]. This side channel is exploited in this work to attack a victim network.

2. **Average Usage/Expected Load:** The typical data traffic or resource demand over time, used to optimize resource allocation and prevent over- or under-provisioning.
3. **Maximum Response Time:** The maximum allowable time to create or modify a slice, which critical for time-sensitive applications such as URLLC.
4. **Slice Duration:** The period during which a slice remains active, which varies depending on service requirements.
5. **Slice Size:** The scope and resources (e.g., bandwidth, storage) allocated to a slice, tailored to meet specific SLAs.

TABLE I
3GPP SPECIFIED VALUES FOR SOME STANDARDIZED 5QIs TO QoS CHARACTERISTICS.

5QI	Averaging Window	Packet Delay Budget	Packet Error Rate	Priority Level	Resource Type
1	2000 ms	100 ms	10^{-2}	20	GBR
2	2000 ms	150 ms	10^{-3}	40	GBR
3	2000 ms	50 ms	10^{-3}	30	GBR
4	2000 ms	300 ms	10^{-6}	50	GBR
5	N/A	100 ms	10^{-6}	10	Non-GBR
6	N/A	300 ms	10^{-6}	60	Non-GBR
7	N/A	100 ms	10^{-3}	70	Non-GBR
8	N/A	300 ms	10^{-6}	80	Non-GBR
9	N/A	300 ms	10^{-6}	90	Non-GBR
65	N/A	75 ms	10^{-2}	7	GBR

B. Long Short-Term Memory Networks (LSTMs)

LSTMs (Long Short-Term Memory networks) are a specialized type of Recurrent Neural Network (RNN) designed to learn and forecast time-dependent behavior of entities. Unlike conventional statistical methods that build probabilistic models based on time-series data (e.g., probability distributions of traffic at each time instant), LSTMs can be trained to predict the likelihood of a traffic peak at any given moment. They do this by analyzing both short-term traffic patterns and long-term trends. LSTMs address common RNN challenges such as vanishing and exploding gradients [41]. Input data – typically time series or text – is usually normalized to the range [0, 1]. Sequences are adjusted to fixed lengths via padding or truncation, and labels are assigned according to the task.

An LSTM model generally consists of three layers: **input**, **dense**, and **output**. The input layer comprises cells that process data through *input*, *forget*, and *output gates*, along with *hidden states* that carry information across time steps and a *cell state* that stores long-term memory [41]. The **dense** layer maps LSTM outputs to the desired output space, while the **output** layer generates final predictions, which may be continuous values or class probabilities [42].

Training LSTMs involves backpropagation through time (BPTT) using optimizers such as **Adam** or **RMSprop** to

adjust weights and biases to minimize loss functions like mean squared error or categorical cross-entropy. This process is repeated over multiple epochs with batch processing [43].

LSTMs are particularly valuable in network contexts, such as 5G, where they process time-dependent data to predict future conditions, detect anomalies, and optimize resource allocation. This enables proactive, real-time network management that enhances performance and security [42]. In this work, LSTMs are also trained to learn and predict peak traffic times in a network to determine an optimal DNS attack window.

III. Overview of the Implemented Simulation Testbed

To demonstrate how 5G network slicing mechanisms can be exploited to launch a DoS attack against a victim network, a simulation-based testbed of a two-site smart manufacturing (Industrial IoT) application using 5G-based networking was developed, as illustrated in Fig. 3. The two sites showcase the creation and management of multiple network slices, each tailored to the specific communication needs of different locations. This allows exploration of the challenges and requirements for secure, efficient communication between branches of a manufacturer engaged in coordinated operations. It also enables evaluation of the impact of a DoS attack on one site and its effect on the other.

The simulation testbed was implemented in OMNeT++, with configurations for both **Site A** and **Site B** detailed in Table II and Table III. **Site A** requires a network slice to support ultra-reliable low-latency communication (uRLLC), with latency under 1 millisecond and a packet error rate no higher than 10^{-5} . To meet these QoS requirements, **Site A** is assigned a high-priority 5G QoS Identifier (5QI) value of 1. The short averaging window helps maintain real-time communication.

Site B, on the other hand, requires a network slice to support massive Machine-Type Communication (mMTC), characterized by massive device connections, short, and low-rate messaging. It can tolerate higher latency and error rates, and is therefore assigned a 5QI value of 9, reflecting its lower priority compared to uRLLC. This slice is designed to handle infrequent data transmissions from numerous devices.

Both sites include sensors, robots, and machines connected via NR modules to the *gNodeB*, as shown in Fig. 3.

Inter-Sites Traffic: Traffic between the two sites is generated by Traffic *TrafficGenerator*. At **Site A**, the generator produces high-priority, low-latency traffic such as control signals for industrial robots and real-time sensor data, which is transmitted to **Site B**. This traffic is routed through **Site A**'s *gNodeB*, forwarded to the VPN Node, and delivered to **Site B**'s network slice via its *gNodeB*.

Conversely, **Site B** generates low-priority, less frequent updates such as machine status reports and maintenance data using its *TrafficGenerator*. These updates are routed through **Site B**'s *gNodeB*, sent through the VPN Node, and delivered back to **Site A**. Traffic coordination is managed by the

NetworkController, which monitors network flows and ensures both sites operate at expected performance metrics.

The RAN: The *gNodeB* module simulates a 5G base station, handling radio communications, managing radio resources, and facilitating data transmission in the 5G network simulation. It adheres to 3GPP standards and is configured in Simu5G for realistic behavior. Setting *nodeType* = **GNODEB** and *NIC-type* = **NRNicEnb** enables 5G-specific functionalities, with connections to the core via the *UPF*. Transmission power is set to the default 100mW. The *gNodeB* supports QoS and network slicing, handling different service types like uRLLC and mMTC. The *N2* and *N3* interfaces connect the *gNodeB* to the 5G Core, managing signaling and data traffic. The *gNodeB* specifications are:

- **Total Bandwidth Available:** 100 MHz (limits overall network capacity).
- **Maximum Supported Users** (maximum number of connected devices per sector): 1,000 users per sector.
- **Number of Sectors** (how coverage area is divided): 2.
- **Maximum Power Output per sector:** 120 mW.
- **Total Resource Blocks**, RBs, (smallest units of radio resources assigned to users): 273 RBs per 20 MHz, totaling 1,365 RBs for 100 MHz.

The Core Network setup parameters for both sites are shown in Table IV. A **network controller** was implemented in Python and integrated into OMNeT++ using Pybind11. It oversees network slicing, and monitors and maintains the networks of different clients to ensure performance and reliability.

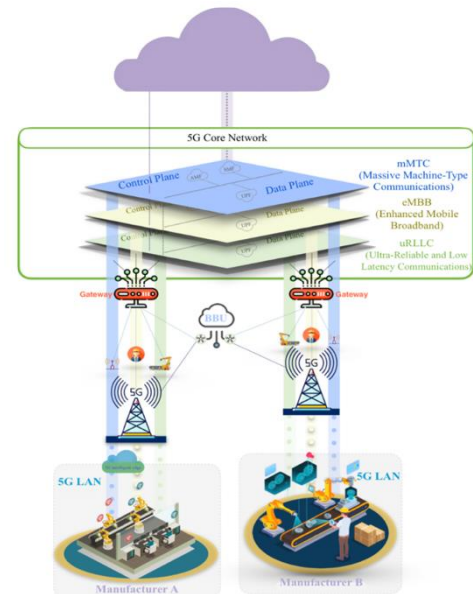


FIGURE 3. Testcase used to simulate a 5G-based smart manufacturer and setting up a network slice by a UE.

Network Slicing: To simulate network slicing in OMNeT++, we adopted an approach that separates and configures the *gNodeB*, *UPF*, *SMF*, and *AMF* modules to

support distinct network slices. This setup ensures that each slice can be independently configured with specific parameters such as bandwidth, QoS, and traffic characteristics. All configurations are defined in the INI configuration file, which specifies the parameters for each network slice. This method enables the testing and analysis of various scenarios without modifying the source code, allowing for efficient management of multiple network slices simultaneously within the simulation – closely aligning with the architecture of 5G-based networks.

The existing predefined modules for *gNodeB*, UPF, SMF, and AMF are parameterized to support multiple slices, facilitating slice-specific traffic management and resource allocation. Network slicing parameters are configured within these modules across the network to ensure that each slice meets its respective QoS requirements and service type, effectively accommodating diverse traffic demands.

TABLE II
MANUFACTURER'S SITE A SIMULATION PARAMETERS.

Parameter	Value	Description
numMachines	100	# of machines
numSensors	50	# of sensors
numRobots	20	# of industrial robots
ue.networkSlice	uRLLC	Network slice type
cellSelection.gnbName	ManufacturerA.gnb	Cell selection based on <i>gNodeB</i> name
*.sendInterval	uniform(1s,5s)	Communication Send Interval
*.useMEC	TRUE	Site A uses MEC
5QI	1	5G QoS Identifier

TABLE III
MANUFACTURER'S SITE B SIMULATION PARAMETERS.

Parameter	Value	Description
numMachines	500	# of machines
numSensors	1000	# of sensors
numRobots	200	# of industrial robots
ue.networkSlice	mMTC	Network slice type
cellSelection.gnbName	ManufacturerB.gnb	Cell selection based on <i>gNodeB</i> name
*.sendInterval	uniform(1s,5s)	Communication Send Interval
*.useMEC	FALSE	Site B uses MEC
5QI	9	5G QoS Identifier

The network simulation setup, illustrated in Fig. 4, integrates multiple modules, including both custom-built components and imported modules from the INET [51] and Simu5G [52] libraries. Together, these modules simulate a 5G-based smart manufacturing environment with dedicated network slices for uRLLC and mMTC services. The *gNodeBs* manage radio resources, while the core network handles distinct traffic flows based on their QoS parameters. A custom-built **network controller** ensures that traffic from

each slice meets performance expectations, and communication between the two sites is efficiently coordinated through VPN and core network routing. Uninfected IoT-23 traffic traces [44] were used to model traffic patterns between IIoT devices at **Sites A and B**.

TABLE IV
CORE NETWORK PARAMETERS USED FOR SIMULATIONS AT BOTH SITES.

Manufacturing site A	Manufacturing site B
Network slice	
Slice type: uRLLC Allocated Bandwidth: 20 MHz Maximum Users: 200 users 5QI: 1	Slice type: mMTC. Allocated Bandwidth: 10 MHz Maximum Users: 2000 devices 5QI: 9
QoS Characteristics	
Averaging Window: 2000 ms Packet Delay Budget: 100 ms. Packet Error Rate: 10^{-2} Priority Level: 20 (Very high) Resource Type: Guaranteed Bit Rate	Packet Delay Budget: 300 ms Packet Error Rate: 10^{-6} Priority Level: 90 (Low to Moderate) Resource Type: Non-Guaranteed Bit Rate

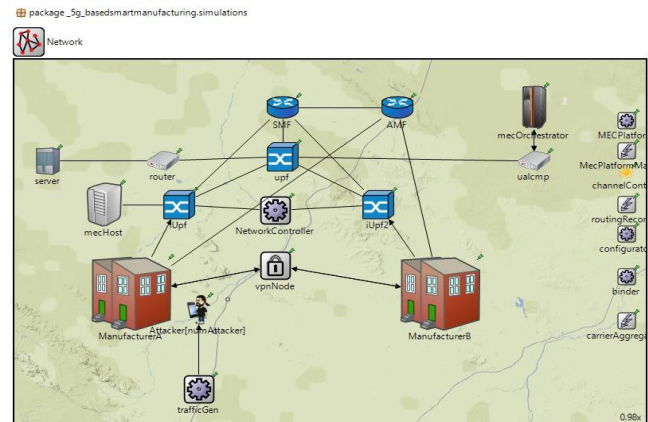


FIGURE 4. Simulation environment setup.

IoT-23 Dataset

The IoT-23 dataset [44] contains real-world traffic captured from IoT devices, encompassing both normal and malicious traffic. It supports packet-level and flow-level analysis, with over 320 million labeled flows making it highly suitable for supervised machine learning applications. In our simulation, we only used uninfected traffic and customized the dataset by segmenting it into two traffic profiles: one for URLLC and another for mMTC. This allowed us to simulate distinct traffic patterns aligned with the specific requirements of each network slice.

Unique IP addresses were assigned to each device to enable precise traffic mapping and routing. Custom traffic generator modules were developed to process the segmented data. This approach ensured that traffic flows were accurately mapped to individual devices, with data scheduled and processed according to realistic communication patterns derived from the dataset.

IV. Proposed Network-Slicing-Based DoS Attack

According to the 5G Network Slicing specifications [35], when a user is granted a large network slice from the same RAN site (*gNodeB*), the available resources for new slices requested by other users on that *gNodeB* are significantly reduced. For example, although the URLLC slice has a high priority level (Priority Level 20), the *gNodeB* may fail to fully satisfy its QoS guarantees if another user is granted a large slice that consumes most of the available resources even while maintaining the requested average traffic.

In the case of **Site A**'s URLLC slice, the *gNodeB*'s total bandwidth is limited to 100 MHz, with 1,365 resource blocks (RBs) shared among all users. If a large slice request consumes a substantial portion of these resources, the remaining bandwidth and RBs available for **Site A**'s URLLC slice are drastically reduced. This leads to several performance issues, including increased latency, difficulty meeting the 100ms packet delay budget (PDB), and potentially higher packet error rates due to network congestion. Under such conditions, the URLLC slice may lose its low-latency and high-reliability guarantees. The *gNodeB* might be unable to maintain the required 20 MHz of bandwidth for **Site A** within the specified averaging window, resulting in delays and unreliable communication for time-sensitive operations.

Additionally, the *gNodeB*'s power limitations (120mW per sector) and user capacity constraints may further strain resources, leading to reduced signal strength or even disconnections for devices relying on the URLLC slice. This degradation can affect the Guaranteed Bit Rate (GBR) provisioned for Site A, resulting in lower throughput and reduced reliability.

In essence, a large slice request with tailored peak traffic designed to maintain the SLA average traffic within the averaging window could be used to launch a DoS attack. This would indirectly cripple the network's ability to provide stable and timely communications to **Site A**. In extreme cases, such overload might trigger admission control mechanisms that reject new connections or degrade the service quality of existing ones, severely impacting the performance of critical URLLC traffic.

This scenario highlights a new potential side-channel attack on network slicing. An attacker can cripple a victim network by requesting a large slice just before peak traffic instances when the victim network would need to request a larger slice. Although this may not appear to be a direct attack, it exploits network standards and specifications to undermine stability and performance. We illustrate how a DoS attack can be launched against an allocated network slice using the phases described below.

It is important to note that this vulnerability exists despite 5G's slice isolation mechanisms, which are enforced only after slices have been granted. The proposed attack prevents the victim network from acquiring the necessary service-level slice in the first place, effectively circumventing slice

isolation entirely. Moreover, the attack does not interfere with the victim network's packets in any way, making it undetectable by firewalls.

The complete workflow of the proposed attack framework is summarized in Fig. 5. The attack consists of five sequential phases: reconnaissance, data acquisition, LSTM-based training, prediction, and attack execution. This implementation was developed in Python 3.9 and integrated into the OMNeT++ simulation framework via the Pybind11 API. In the reconnaissance phase the attacker observes slice allocation responses within the OMNeT++-based 5G environment. The collected side-channel data are logged and used to train a LSTM model to predict future peak traffic windows. Once a potential peak period is detected, the attacker generates high-bandwidth slice requests just before that window, leading to slice resource saturation and QoS degradation. This sequential process ensures replicability and highlights how predictive analytics can be leveraged to exploit 5G network slicing mechanisms.

Below is the detailed description of the different phases of this attack.

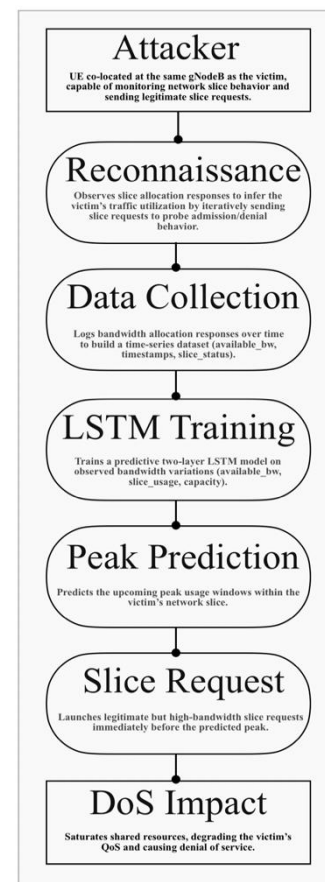


FIGURE 5. Workflow of the Proposed Network-Slicing-Based DoS Attack Framework.

A. Reconnaissance Phase

In this phase, network slicing mechanisms are used to gather

information about the victim network's traffic patterns over time via the side-channel method described above. Assuming the attacker is connected to the same *gNodeB* as the victim network (**Site A**), they request various network slices with different capacities over several days at regular intervals of 2 seconds matching the evaluation cycle of the victim's averaging window (assumed to be 2000 ms). The flow chart of the mechanism implemented in OMNeT++ is shown in Fig. 6.

By analyzing the network controller's responses (granting or denying the requested slice capacity) and observing how the network reacts to varying traffic loads, the attacker can build a detailed traffic profile (bandwidth vs. time) of the victim network. The 5QI slices used for reconnaissance either a single slice or multiple simultaneous slices with increasing bandwidth allocations to exploit the remaining 70 MHz of bandwidth are shown in Table V. The attacker tests different priority levels and delay budgets, expanding their reconnaissance scope and probing the network with bandwidth demands ranging from 10 MHz to 40 MHz, while monitoring its behavior.

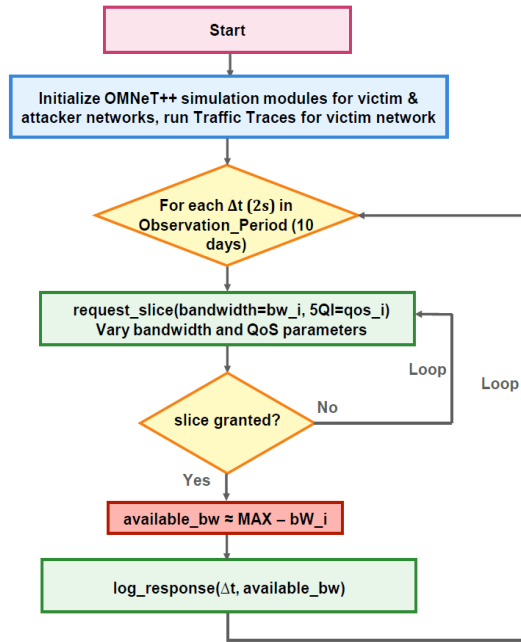


FIGURE 6. Flow chart of the OMNeT++ implementation of the reconnaissance phase.

TABLE V
THE 5QIS USED FOR RECONNAISSANCE PHASE OF THE ATTACK.

5QI	Bandwidth	PDB	Priority Level	Resource Type
6	10 MHz	300 ms	60	Non-GBR
7	35 MHz	100 ms	70	Non-GBR
8	20 MHz	300 ms	80	Non-GBR
9	40 MHz	300 ms	90	Non-GBR
10	30 MHz	1100 ms	90	Non-GBR

B. Building The Predictive LSTM Traffic Model

The data collected during the reconnaissance stage is used to train an LSTM model such that it can predict network traffic peaks in the victim network. Extensive experiments were conducted to optimize the LSTM architecture using training data obtained from the reconnaissance phase, which was implemented within the simulation framework. The final optimized model consists of two LSTM layers. The first contains 128 memory cells with return sequences enabled (True), allowing outputs to be passed to the next layer. The second layer has 64 memory cells with return sequences disabled (False). A fully connected dense layer with a single neuron is used for binary output (peak time prediction), employing a Sigmoid activation function to classify whether the next time step corresponds to a traffic peak.

To mitigate overfitting, a dropout rate of 20% is applied after each LSTM layer. The model is trained using the Adam optimizer with a learning rate of 0.001, and binary cross-entropy is used as the loss function to predict whether traffic is at its peak (1) or not (0). Training is performed with a batch size of 128 over 20 epochs.

Throughput, latency, and packet-loss were derived from OMNeT++ vector logs and averaged over fixed 10-second intervals to minimize short-term fluctuations. After several exploratory runs, the LSTM hyperparameters were fine-tuned to achieve stable convergence. Key hyperparameters including the number of layers, learning rate, and dropout rate were empirically optimized using a grid search to minimize validation loss. To verify robustness, the model was retrained across five random seeds, exhibiting consistent convergence behavior and stable prediction accuracy (MAE = 0.037 ± 0.004). These results confirm that the selected configuration generalizes well under varying traffic conditions. Fig. 7 illustrates the LSTM model architecture, and Table VI summarizes the model parameters. The model uses a window of 50 timesteps (i.e., 50 sequential data intervals) for prediction, incorporating three features: Estimated Traffic Volume, Utilized Slice Bandwidth, and Slice Capacity.

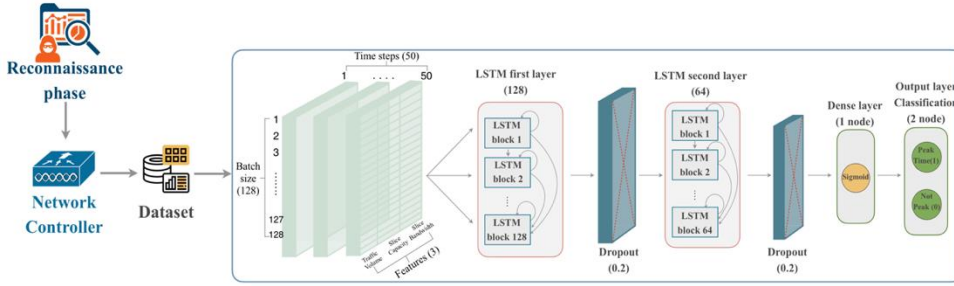


FIGURE 7. The utilized LSTM model architecture.

C. Launching the Attack

During the actual attack, the attacker applies the same reconnaissance technique to collect information about the victim network's traffic patterns. These are fed to the trained LSTM model to predict upcoming peak traffic times. The attack is subsequently launched by requesting large network slices just before these predicted peaks, flooding the network within the averaging window and causing a significant degradation in the victim network's performance.

V. Experimental Setup and Environment

The experiments were conducted using OMNeT++ 6.0 simulations integrated with INET and Simu5G frameworks to emulate a 5G-based smart manufacturing environment consisting of two industrial sites. Each site was configured with distinct network slices representing uRLLC and mMTC traffic, generated using IoT-23 dataset traces. The CSV-based traces were applied as UDP applications to simulate realistic communication patterns for industrial devices and robotic agents.

TABLE VI
SUMMARY OF THE LSTM MODEL TRAINING PARAMETERS.

Parameter	Value
Window of Timesteps	50 timesteps
Number of Features	3 (Estimated Traffic Vol., Utilized Slice Bandwidth and Capacity)
Training Size	80% of dataset
Test Size	20% of dataset
Number of LSTM Layer	2
LSTM Layer 1 Units	128 units
LSTM Layer 2 Units	64 units
Dropout Rate	0.2 (20%)
Dense Layer Units	1 unit
O/P Activation Function	Sigmoid
Loss Function	Binary Cross-Entropy
Optimizer	Adam
Batch Size	128
Epochs	20

The simulation testbed comprised a virtualized 5G core network, radio access modules, and MEC hosts configured as simple and compound modules within OMNeT++. The

attacker node was deployed within Manufacturer A's radio range and configured to monitor slice response behavior during the reconnaissance phase. The attacker node collected side-channel information such as slice bandwidth allocation and response times, which were later used to train a two-layer LSTM model for peak traffic prediction.

All simulations were executed on a Windows 11 workstation (Intel Core i7 CPU, 32 GB RAM). Multiple configurations with varying bandwidth and priority parameters were tested to confirm consistent behavior. The LSTM-based prediction model was validated by comparing the predicted peak time windows with the actual traffic variations observed in the simulated environment. Network degradation was analyzed qualitatively, as the predicted peaks coincided with noticeable reductions in slice throughput and increased queuing delay during the attack execution. The simulation parameters and random initialization were fixed across all runs to ensure consistent and reproducible behavior. The observed outcomes confirm that the proposed framework can effectively anticipate high-traffic periods and trigger resource contention leading to QoS degradation.

To strengthen the analysis, statistical comparisons were performed between the baseline and attack periods for key metrics, throughput, latency, and packet loss. The mean throughput decreased by more than 90 %, while average latency and packet loss increased by over 200 % and 150 %, respectively. A two-sample t-test confirmed that these differences are statistically significant ($p < 0.01$), indicating that the observed degradation is a direct consequence of the proposed attack rather than natural traffic fluctuations. To ensure numerical stability and reproducibility, all feature values were normalized to the $[0,1]$ range, and random seeds were fixed across simulation runs. Each scenario was repeated three times under identical configuration parameters, producing consistent results with negligible variation.

VI. Experimental Results

A. Simulation Setup

In OMNeT++, the attacker module is modeled as a UE under the NR framework and is named *Attacker*. The simulation configuration is implemented in the INI file using the same

approach applied to network slicing. Multiple slice request scenarios are defined for the *Attacker*, where it requests slices at regular intervals every 2 seconds, resulting in a total of 50 requests over a 100-second simulation period to simulate the reconnaissance phase.

During this phase, the *Attacker* module requests network slices with varying parameters such as bandwidth, priority, and packet delay budget to gather data on network peak times, slice capacities, and other critical metrics. The *Attacker's* module logs and captures the responses to each request, identifying peak traffic times (i.e., when available capacity is at its minimum) and slice capacities. The collected data includes: Estimated Traffic Volume (Mbps); Slice Bandwidth (MHz); Packet Delay (ms); Slice Capacity (%); and Available Bandwidth (MHz).

Each data point is timestamped to preserve the sequential nature of the victim network's traffic changes over time.

B. Implementation of the Reconnaissance Phase

The reconnaissance phase was implemented using Python scripts within OMNeT++. Fig. 8 below presents the simulation results of this phase, showing the estimated *available* network bandwidth (in MHz) compared to the *actual* bandwidth utilized by the victim network over a 2-minute traffic period. It demonstrates how the attacker can infer the victim's traffic patterns solely by observing slice admission responses over time.

Fig. 9 shows the difference between the attacker's estimation of the *available* network bandwidth versus the *actual available* bandwidth during the same period.

Although the attack tends to overestimate the available bandwidth, it accurately identifies the time instances of the victim's traffic peaks those moments when available bandwidth is at its minimum. This overestimation is not a critical issue, as it implies the attacker would require a smaller slice than estimated to effectively disrupt the victim's traffic.

C. Training of the LSTM Model

The LSTM model's output is a binary classification for each time step, predicting whether the upcoming 2-second time slot contains a traffic peak. During the training phase, labeling of the traffic data (estimated from the reconnaissance phase) is based on a simple statistical model. The collected traffic distribution is fitted to a Gaussian model. Two criteria are then used to label peak periods: either a threshold of 90% of the maximum traffic value or a value within 1.5 standard deviations from the maximum.

Fig. 10 shows the actual victim network's traffic over a work-day period (6 AM to 5 PM), highlighting the ideal attack window at a 140 Mbps peak. Fig. 11 presents the actual attack window as predicted by the LSTM model.

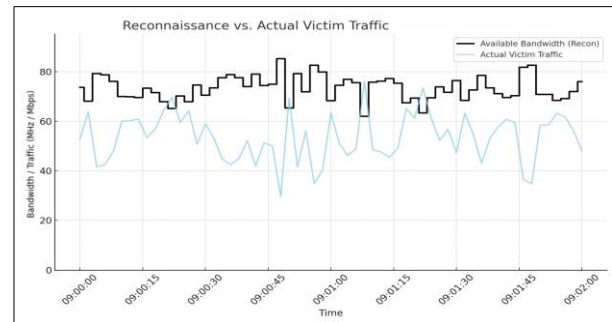


FIGURE 8. The attacker's view (black line) of the available bandwidth, as observed during the reconnaissance phase, compared to the actual traffic sent by the victim (blue line).

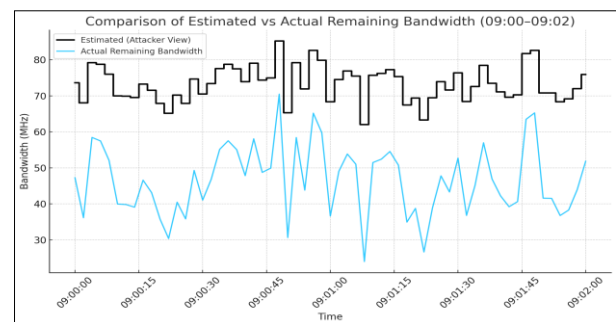


FIGURE 9. Comparison between the attacker's estimated available bandwidth (black line) and the actual available bandwidth (blue line).

D. The DoS Attack Phase

The final phase of the simulation in OMNeT++ models the attack by having the *Attacker* acquire multiple 5QIs with a large total bandwidth just before the victim's peak traffic, as predicted by the LSTM model. The total RAN bandwidth at the *gNodeB* is 100 MHz, which is dynamically allocated based on demand i.e., no fixed portion is reserved in advance for any particular slice, including URLLC. Typically, 10 MHz is used for control and signaling, leaving 90 MHz dynamically available for allocation.

Knowing that **site A** requires a 20 MHz URLLC slice, the attacker exploits the dynamic bandwidth allocation by requesting a slice of 75 MHz bandwidth or multiple slices with a total bandwidth of 75 MHz, such as those using 5QI7 and 5QI9, just before the predicted peak times of **Site A**. This prevents **Site A** from receiving its required URLLC slice, disrupting the critical services it depends on. Meanwhile, the attacker's slices remain unaffected due to their isolated and guaranteed allocations. As a result, the *gNodeB* control plane begins rejecting further URLLC slice requests. No new slices or expansions can be initiated until overall demand decreases, existing slices are scaled down, or additional resources become available. This leads to significant service degradation or even disruption of operations at **Site A**.



FIGURE 10. The actual victim network's traffic over a full work day and the ideal attack window.

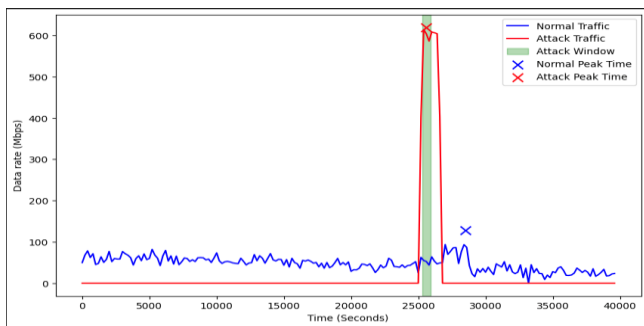


FIGURE 11. The actual launched attack window as predicted by the LSTM model.

In the simulated attack, the *Attacker* achieves this by using a combination of devices to request different slice sizes. For example, one device requests a 40 MHz slice, consuming 546 RBs, while two other devices request 15 MHz slices each, consuming 204 RBs per slice. In total, 954 RBs out of the 1,228 RBs available for dynamic allocation are consumed equivalent to 75 MHz of bandwidth. By taking up this much bandwidth, the attacker ensures that when **Site A** attempts to allocate its 20 MHz URLLC slice, there is insufficient bandwidth left for the network to fulfill the request. Even though URLLC slices are given high priority, the network cannot allocate the required resources (RBs), resulting in severe congestion or a denial-of-service (DoS) for **Site A**.

The **averaging window** further enables the attack by allowing the *Attacker* to gradually consume bandwidth without immediate detection. As previously explained, resources are allocated to network slices based on average traffic over a set time window, rather than responding instantly to traffic surges. The attacker takes advantage of this by distributing multiple slice requests across the averaging window, making the traffic appear normal. By timing these requests just before the victim's peak periods, the attacker pre-allocates a significant portion of the bandwidth. The averaging window conceals the attack until the network is already strained, making it highly effective.

It should be noted that the averaging window was also incorporated into the LSTM training process. This provided insight into how performance metrics such as delay and bandwidth allocation, fluctuate, allowing the model to predict traffic peaks across multiple averaging windows and refine its ability to anticipate the heaviest loads.

Fig. 12 and Fig. 13 show the effects of the attack on the victim network's throughput, packet loss rate, and latency before and during the attack for both sites, as captured from the simulations. These figures illustrate the network parameters before and during the attack. Both throughput and latency at **Site A** were critically affected, as most of the bandwidth at its *gNodeB* was consumed by the attacker. This resulted in a significant reduction in throughput (down to nearly zero) and a dramatic increase in latency and packet loss effectively preventing **Site A** from maintaining its critical real-time operations.

Since **Site B** (which uses an mMTC slice) is connected to a different *gNodeB*, the attack did not directly affect its throughput, latency, or packet loss. However, due to its communication with **Site A**, the congestion at Site A's *gNodeB* indirectly impacted **Site B**'s throughput, causing delays and reduced transmission speed. Packet loss within Site B's own network remained normal, but any communication with **Site A** experienced increased packet loss. These delays disrupted joint operations and coordination between the two branches of the manufacturer.

VII. Potential Defense Mechanisms

Several potential mitigation strategies can be employed to detect the proposed attack. The most notable of those, is anomaly-based slice-request monitoring applied at the network-slice manager. This could itself be an LSTM model that detects irregular or high-frequency slice allocation behaviors that deviate from normal admission patterns. Attackers could attempt to defeat such a measure by deploying multiple virtual networks and coordinating the slice-requests from multiple devices on these networks to defeat the detection model. To counteract that, network operators could add adaptive/dynamic admission-control mechanisms to adjust slice resource assignments based on real-time congestion prediction, preventing resource starvation during coordinated attacks. Future research could integrate these defensive measures with intelligent monitoring agents to establish a resilient and self-healing slicing architecture.

VIII. Validity Threats and Limitations

As all experiments were conducted using simulation tools, certain validity threats must be considered. Internal validity may be affected by the assumptions and simplifications inherent in the OMNeT++/Simu5G environment. Although these frameworks accurately model 5G core and RAN behaviors, factors such as radio interference, background noise, and real hardware scheduling were abstracted for

tractability. Nevertheless, all parameters including bandwidth, QoS profiles (5QI), and latency budgets were derived from 3GPP specifications to preserve realism and ensure that the observed effects genuinely reflect network-slicing dynamics. External validity concerns the generalization of results beyond the simulated scenario. The two-site industrial setup and the uRLLC/mMTC slicing model represent a common 5G smart manufacturing configuration. However, actual deployments may vary in slice orchestration policies, radio configurations, or resource scheduling algorithms, which could influence the attack's impact. Despite this, the proposed DoS mechanism leverages standardized slicing and resource-allocation behavior, making its underlying principle applicable to real-world 5G networks and potentially other network-sliced domains.

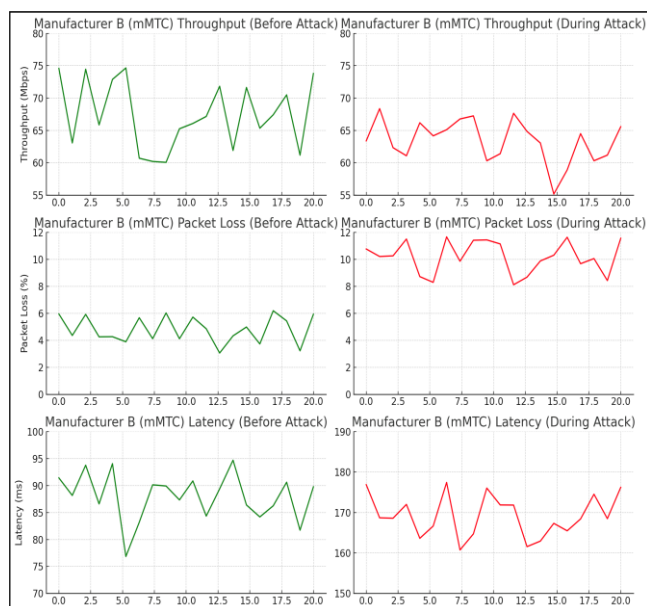


FIGURE 12. Site A's network throughput, packet loss rate, and latency before (left hand side) and during the DoS attack (right hand side).

IX. CONCLUSION

In this work, a DoS attack against 5G core networks using standard 5G network slicing mechanisms was identified and successfully demonstrated. A simulation testbed consisting of two smart manufacturing sites connected to a 5G network was used as a case study and simulated using the OMNeT++ framework. Through this simulation testbed, we demonstrated how a network slice of a manufacturing site can be systematically and subtly targeted in a DoS attack that is difficult for the network to detect.

A side channel based on standard 5G network slicing mechanisms was identified and effectively used to infer the victim's network traffic patterns. An LSTM model was then employed to predict peak traffic times for the victim network and request a large network slice, consuming most of the available resources and bandwidth. This rendered the

network unable to allocate the necessary resources for the victim slice. The exploit leveraged the virtualized and dynamic nature of network slices, as well as the flexible allocation of bandwidth and resources.

Simulation results indicated that the attack successfully deprived the victim network slice of its ability to request essential communication services and indirectly impacted other networks that rely on communication with the targeted slice. These findings highlight how DoS attacks targeting critical network slices can severely disrupt communication and operations in a 5G-enabled smart manufacturing environment even when only one site is directly targeted.

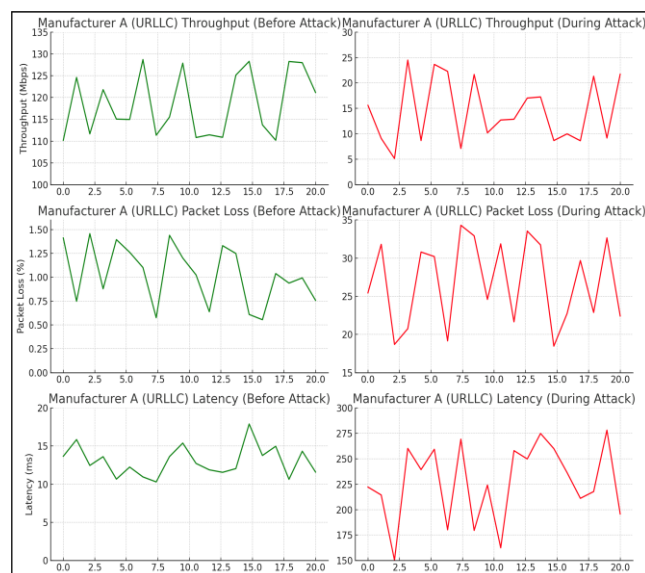


FIGURE 13. Site B's network throughput, packet loss rate, and latency before (left hand side) and during the DoS attack (right hand side).

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